

# Zizhe Zhang

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[website](#) | [github](#) | [linkedin](#) | [scholar](#)

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>University of Pennsylvania, GRASP Lab</b><br>Master of Science in Engineering (M.S.E.) in Robotics<br>Advisor: Prof. Nadia Figueroa                                                | Aug 2024 – Present<br>Philadelphia, PA |
| <b>University of California San Diego, Jacobs School of Engineering</b><br>Visiting Student                                                                                           | Jan 2023 – Mar 2023<br>San Diego, CA   |
| <b>Southeast University, School of Instrument Science and Engineering</b><br>Bachelor of Engineering (B.E.) in Measuring Control Technology & Instruments<br>Advisor: Prof. Yuan Yang | Aug 2020 – Jun 2024<br>Nanjing, China  |

## PUBLICATIONS

(\* and † denote equal contribution)

### Viability-Preserving Passive Torque Control

**Zizhe Zhang\***, Yicong Wang\*, Zhiquan Zhang, Tianyu Li, and Nadia Figueroa  
Under Review

[[pdf](#)] [[Website](#)]

### Flow with the Force Field: Learning 3D Compliant Flow Matching Policies from Force and Demonstration-Guided Simulation Data

Tianyu Li\*, Yihan Li\*, **Zizhe Zhang**, and Nadia Figueroa  
Under Review

[[pdf](#)] [[Website](#)]

### VLMgineer: Vision Language Models as Robotic Toolsmiths

George Jiayuan Gao\*, Tianyu Li\*, Junyao Shi, **Zizhe Zhang†**, Yihan Li†, Nadia Figueroa, and Dinesh Jayaraman  
Under Review at International Conference on Learning Representations (ICLR), 2026  
Robotics: Science and Systems (RSS) Workshop on Robot Hardware-Aware Intelligence (**Oral Spotlight**), 2025

[[arXiv](#)] [[PDF](#)] [[Website](#)]

### Image-Based Visual Servoing for Enhanced Cooperation of Dual-Arm Manipulation

**Zizhe Zhang**, Yuan Yang, Wenqiang Zuo, Guangming Song, Aiguo Song, and Yang Shi  
IEEE Robotics and Automation Letters (RA-L), 2025

[[arXiv](#)] [[PDF](#)] [[Website](#)]

## RESEARCH AND INDUSTRY EXPERIENCE

**Visiting Scholar**, Duke University  
Robot Dexterity Lab, advised by Prof. Xianyi Cheng  
Working on developing an adaptive compliance motion representation for generalizable robotic manipulation and physical human-robot interaction (In Progress)

Durham, NC

Jun 2025 – Aug 2025

**Graduate Research Assistant**, University of Pennsylvania  
Figueroa Robotics Lab, advised by Prof. Nadia Figueroa

Philadelphia, PA

Nov 2024 – Present

Working on machine learning applications in control theory and robot manipulation. Projects include:

- Ensuring feasibility of passivity-based torque control via learned kinematic constraint mappings (*Under Review*)
- Implementing vision-based learning with force feedback for contact-rich manipulation tasks (*Under Review*)
- Utilizing Vision Language Models (VLMs) for robot policy synthesis ([RSS Workshop 2025](#))

**Technical Intern**, Schneider Electric

Shanghai, China

Working in the Kylin project, including:

Jun 2023 – Aug 2023

- IGBT thermal simulations
- Capacitor lifetime assessments
- EMC testing
- Circuit design

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Undergraduate Research Assistant</b> , Southeast University                                                                                                                                                                                                                          | Nanjing, China      |
| <b>Robotic Perception and Control Lab</b> , advised by Prof. Yuan Yang                                                                                                                                                                                                                  | Dec 2023 – Aug 2024 |
| Worked on developing visual-servoing-based robotic control architectures. Projects include:                                                                                                                                                                                             |                     |
| <ul style="list-style-type: none"> <li>Designed an enhanced dual-arm collaborative control system based on image-based visual servoing (<a href="#">RA-L 2025</a>)</li> <li>Designed a shared teleoperation system control based on visual servoing (<i>Bachelor Thesis</i>)</li> </ul> |                     |
| <b>Automotive Safety Technology Lab</b> , advised by Prof. Dong Wang                                                                                                                                                                                                                    | Mar 2022 – Jun 2023 |
| <ul style="list-style-type: none"> <li>Applying edge detection to classify Martian topography and identify soft-ground hazards</li> <li>Designed a wheeled-leg ground-detection mechanism for rover traversal and analyzing force signals to predict mobility</li> </ul>                |                     |
| <b>Advanced Navigation Technology Lab</b> , advised by Prof. Xuanpeng Li                                                                                                                                                                                                                | Jul 2023 – Sep 2024 |
| <ul style="list-style-type: none"> <li>Utilizing the PYTS library for signal visualization and building a CNN to extract and classify features from ADS-B radio signals</li> </ul>                                                                                                      |                     |

COURSE PROJECTS

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>News Source Classification</b> , University of Pennsylvania                                                                                                                                                                                                                                                                                                                                                                                                        | Philadelphia, PA    |
| <i>Team of 3, Leader</i>                                                                                                                                                                                                                                                                                                                                                                                                                                              | Aug 2024 – Dec 2024 |
| <ul style="list-style-type: none"> <li>Collected and preprocessed 415,343 headlines from Fox News and NBC using sitemap scraping and BeautifulSoup, ensuring a balanced, cleaned dataset</li> <li>Built a TF-IDF + Logistic Regression baseline and fine-tuned BERT and DeBERTa models, applying k-fold cross-validation to eliminate data leakage</li> <li>DeBERTa K-fold achieved 88.66 % test accuracy, significantly surpassing the baseline (67.61 %)</li> </ul> |                     |
| <b>Autonomous Vehicle based on GPS &amp; DoF Camera</b> , University of California San Diego                                                                                                                                                                                                                                                                                                                                                                          | San Diego, CA       |
| <i>Team of 3</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jan 2023 – Mar 2023 |
| <ul style="list-style-type: none"> <li>Utilized Python and VESC to control the robot, DoF camera to find and track lanes, centimetric GPS and PID method to record and follow paths</li> <li>Brought the robot to a complete stop by using PyVesc and DepthAI libraries to run stop sign detection on the camera</li> <li>Enabled the robot to respond correctly to speed limit signs by performing text detection on the camera</li> </ul>                           |                     |
| <b>Design and Implementation of a Weather Query and Display Module</b> , Southeast University.                                                                                                                                                                                                                                                                                                                                                                        | Nanjing, China      |
| <i>Team of 3, Leader</i>                                                                                                                                                                                                                                                                                                                                                                                                                                              | Jan 2023 – Mar 2023 |
| <ul style="list-style-type: none"> <li>Used ASM to design the function and organize each part of the MCU. Designed a Weather Query and Display Module which can choose and display weather information of several customized cities</li> <li>The whole project is based on a small device which can be used as smart home device</li> </ul>                                                                                                                           |                     |

HONORS AND AWARDS

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|-------------------------------------------------------------|------|
| <b>IROS-SDC travel award</b> , Committee of IROS 2025       | 2025 |
| <b>Outstanding Bachelor’s Thesis</b> , Southeast University | 2024 |
| <b>Merit Student</b> , Southeast University                 | 2021 |

SERVICE

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|--------------------------------------------------------------------------------------------|-----------|
| <b>Reviewer</b>                                                                            |           |
| IEEE Transactions on Robotics (TRO)                                                        | 2025      |
| IEEE Transactions on Industrial Electronics (TIE)                                          | 2025      |
| IEEE Transactions on Industrial Informatics (TII)                                          | 2025      |
| <b>Teaching Assistant</b>                                                                  |           |
| ESE 5000 Linear Systems Theory, University of Pennsylvania, advised by Prof. George Pappas | Fall 2025 |
| <b>Volunteer</b>                                                                           |           |
| Robotics: Science and Systems (RSS)                                                        | 2025      |

SKILLS

|                    |                                                                                   |
|--------------------|-----------------------------------------------------------------------------------|
| <b>Programming</b> | C, C++, Python, MATLAB                                                            |
| <b>Tools</b>       | CMake, PyTorch, ROS, RLBench, ManiSkill, CoppeliaSim, PyBullet, Issac Gym/Sim/Lab |
| <b>Robots</b>      | UR3, UR3e, Franka Emika Panda, Franka Research 3                                  |